

Data-Driven Insights into Competitive Electricity Supply Participation in Massachusetts



Massachusetts Department of Energy Resources (DOER) | 2020–2025

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Business Problem

Market Context

Massachusetts operates a partially deregulated electricity market. Customers may choose between their incumbent utility provider and alternative competitive electricity suppliers.

This raises critical questions: who switches, who stays, and why?

The Project Goal

- Assess the differences in electricity consumption patterns across customer classes.
- Analyze the customer segments that are most likely to adopt competitive electricity supply
- Determine the relationship that exists between usage volume and supplier choice
- Evaluate the trends and changes in the competitive market participation over time between 2020–2025

Customer Segments Analyzed

-  **Residential**
Standard household accounts
-  **Small C&I ($\leq 3,000$ kWh/mo)**
Small businesses & retail
-  **Medium C&I (3K–120K kWh/mo)**
Mid-size commercial
-  **Large C&I ($> 120,000$ kWh/mo)**
Industrial & major commercial
-  **Farm**
Agricultural accounts
-  **Streetlights**
Municipal infrastructure

Data Nuances

Data Preparation

What was done to make the data usable

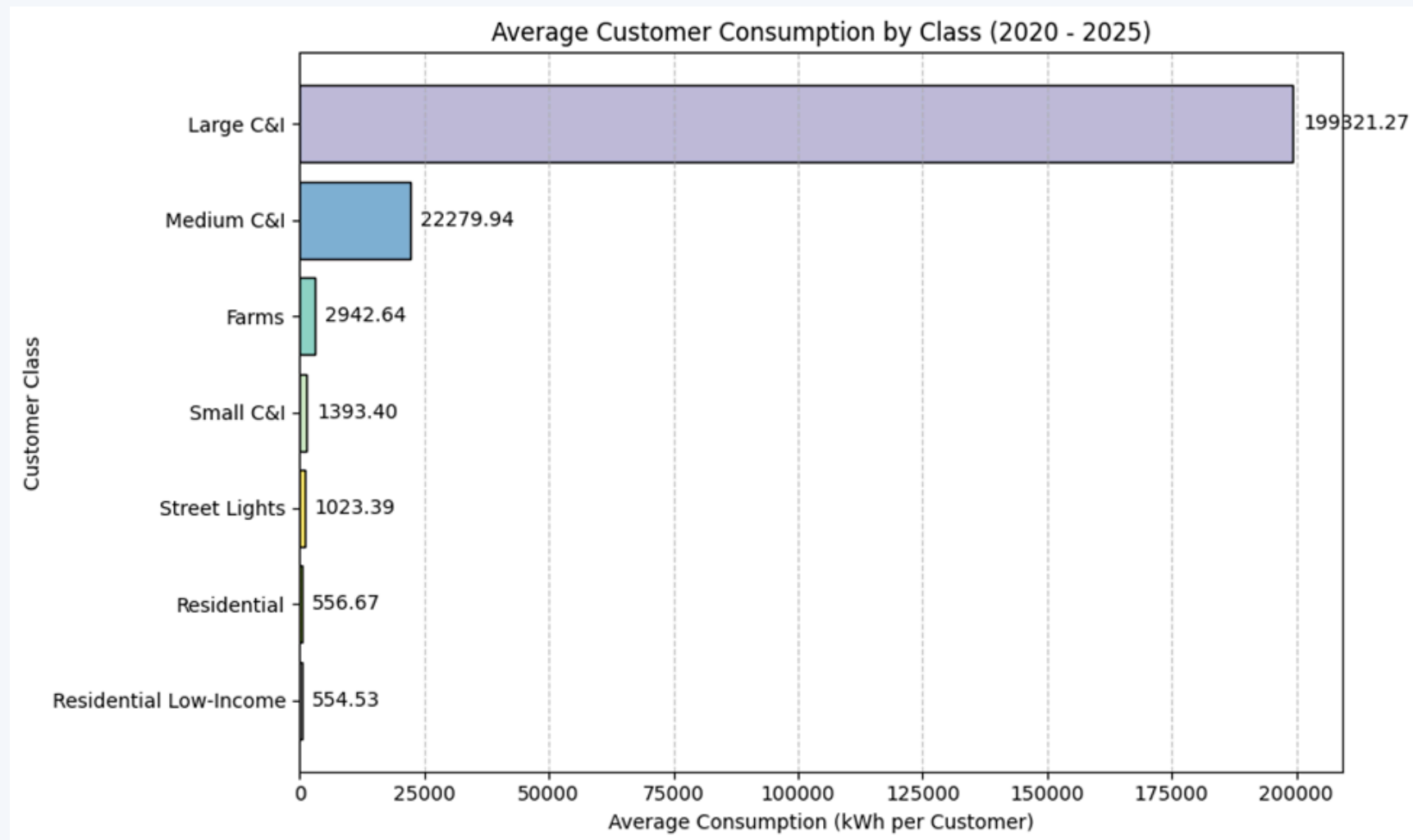
- **Flatten nested headers:** Multi-level merged layout reshaped into flat analytical format.
- **Subtotal rows removal:** excluded aggregated rows to prevent double-counting.
- **Established Color Palette:** for various customer class categories to maintain visual clarity across all charts.
- **Normalized the machine learning model:** by applying a logarithmic transformation to mitigate skewness.

Data Limitation

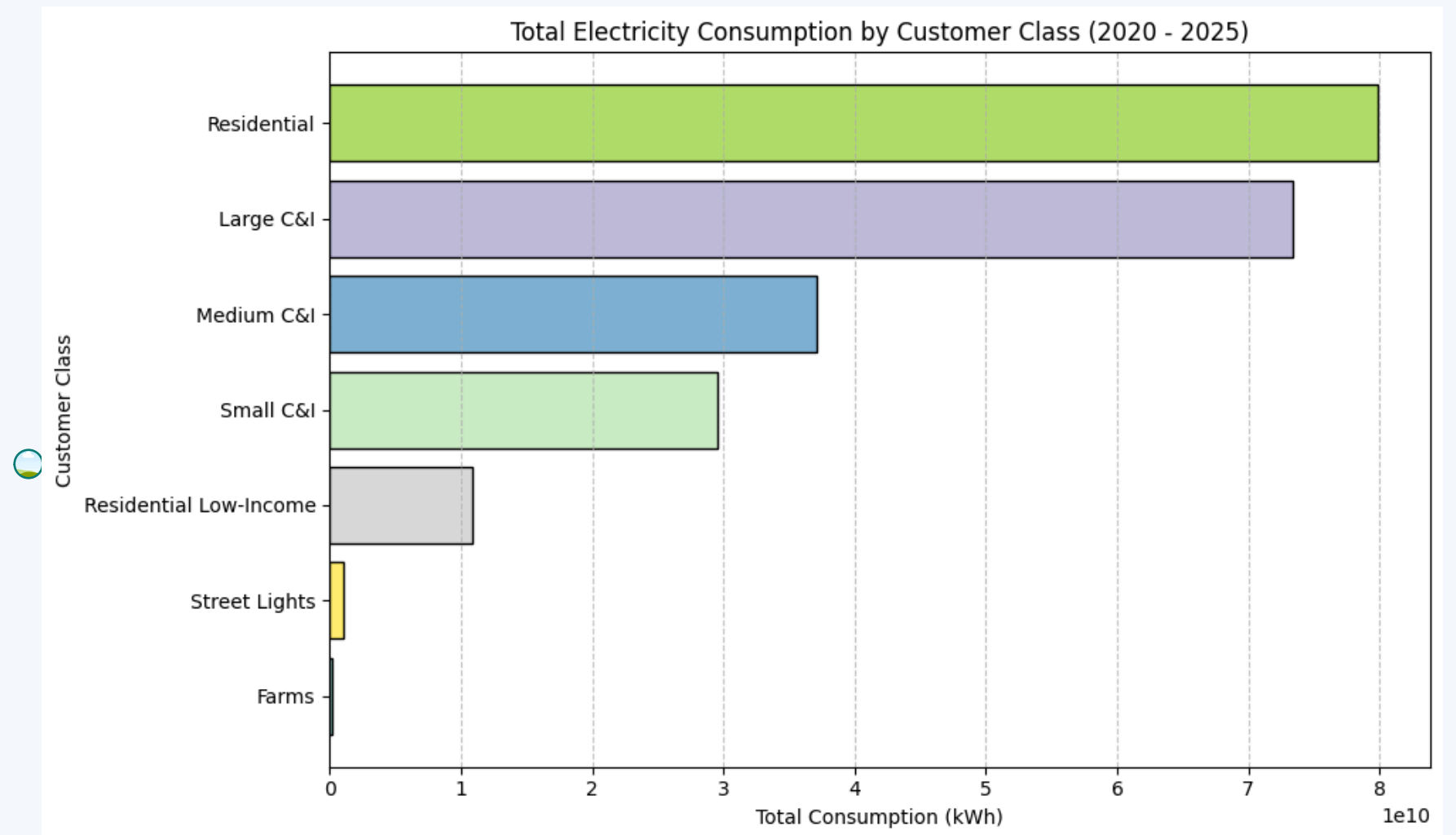
The irregularities we changed and other times, managed

- ⚠ **Farm rows data inconsistency:** National Grid and utilities show null values across all Farm rows.
- ⚠ **Null Municipal Aggregation data:** for most of 2020 and parts of 2021.
- ⚠ **Residential Low-Income:** treated as part of the general Residential segment.
- ⚠ **Class Imbalance and underrepresentation:** of the basic service, leading to a skewed test set.

Key Findings: Average consumption differs from total consumption

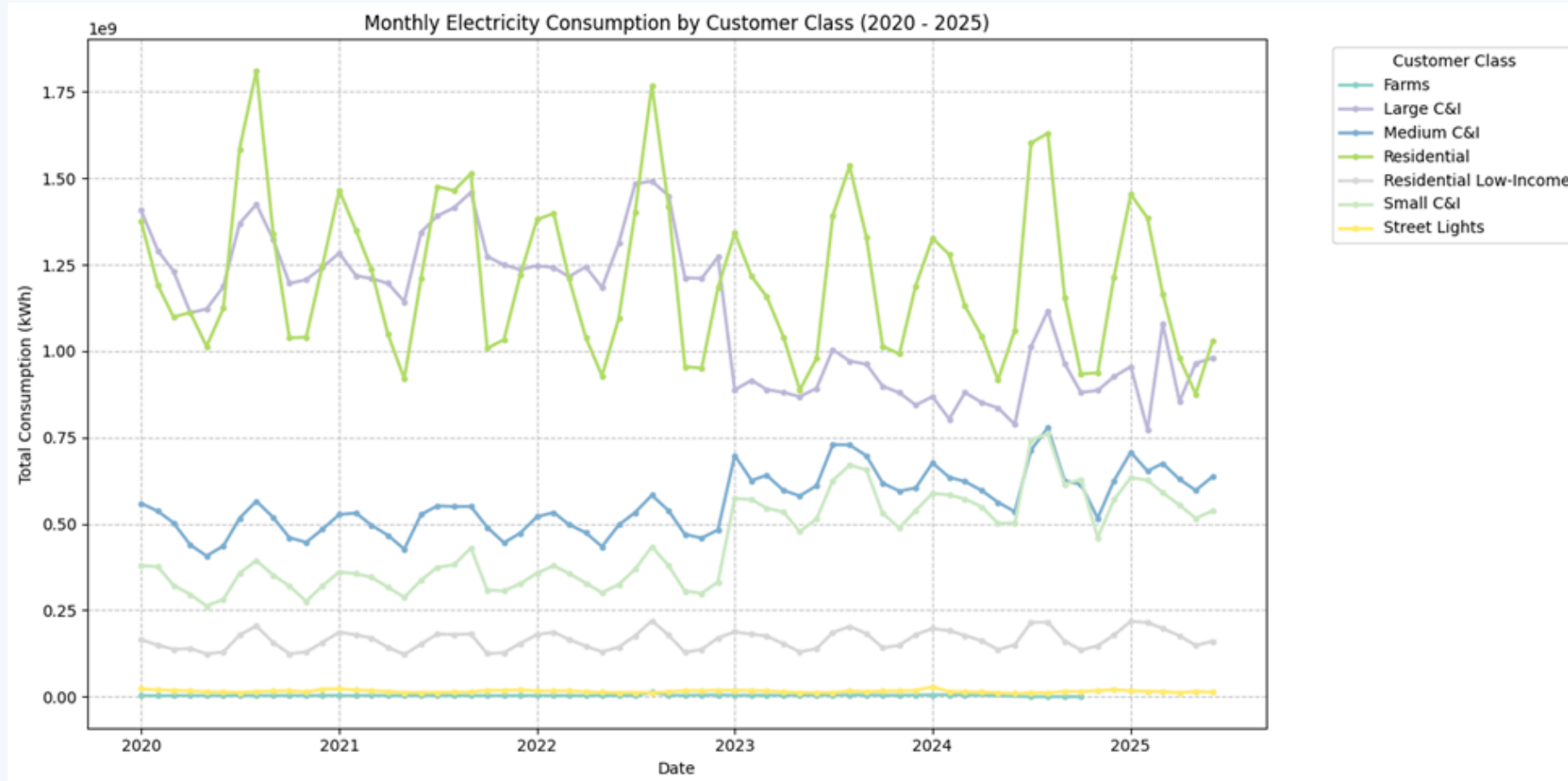


Large C&I customers consume an average of 199,000 kWh per month per customer, compared to just 557 kWh for Residential. This 350x difference highlights the extreme consumption concentration in a small number of industrial accounts.

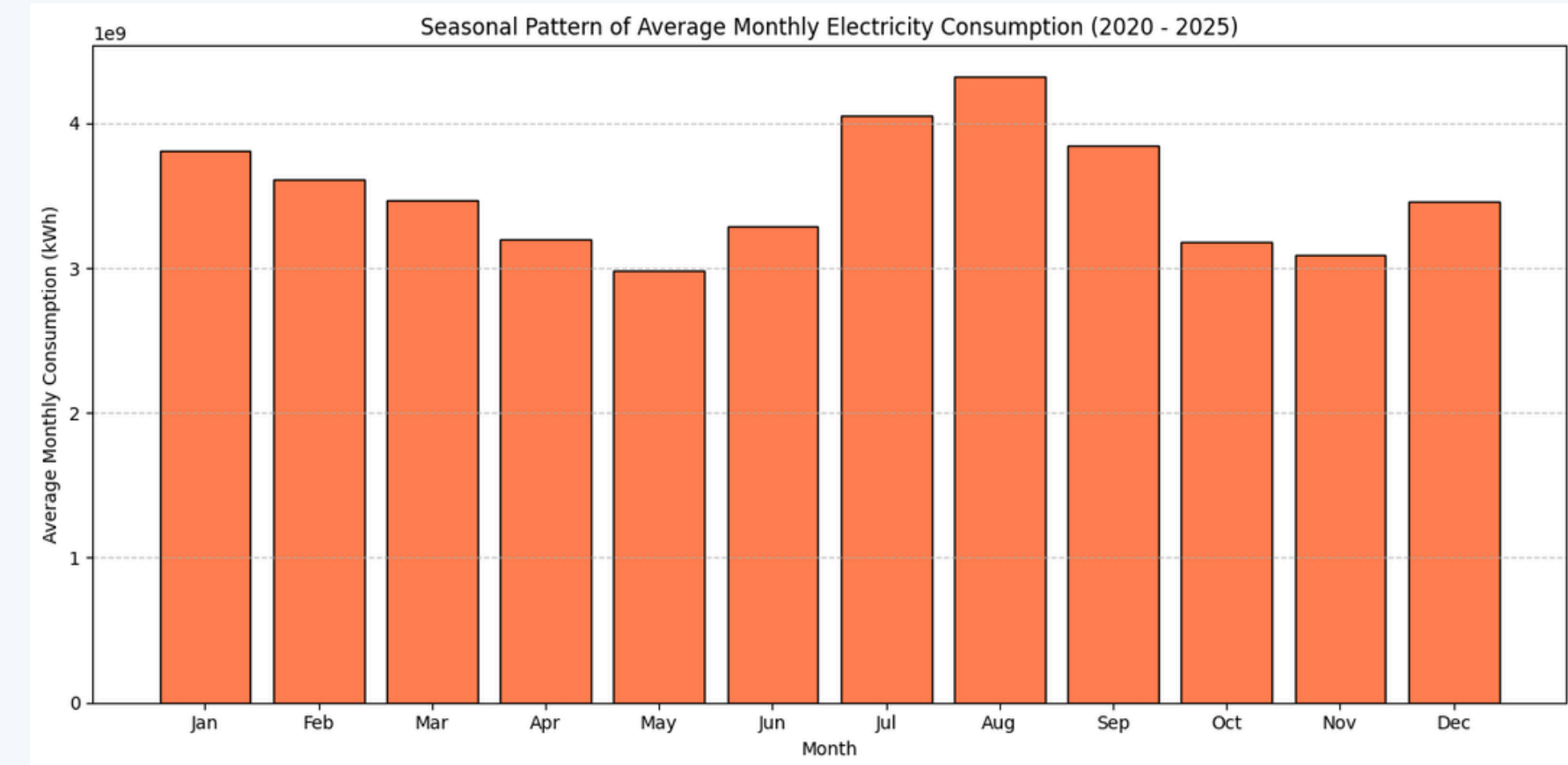


Residential dominance in total consumption is driven entirely by the sheer volume of residential accounts and not by individual customer usage

Key Findings: Seasonal pattern differs across customer segments

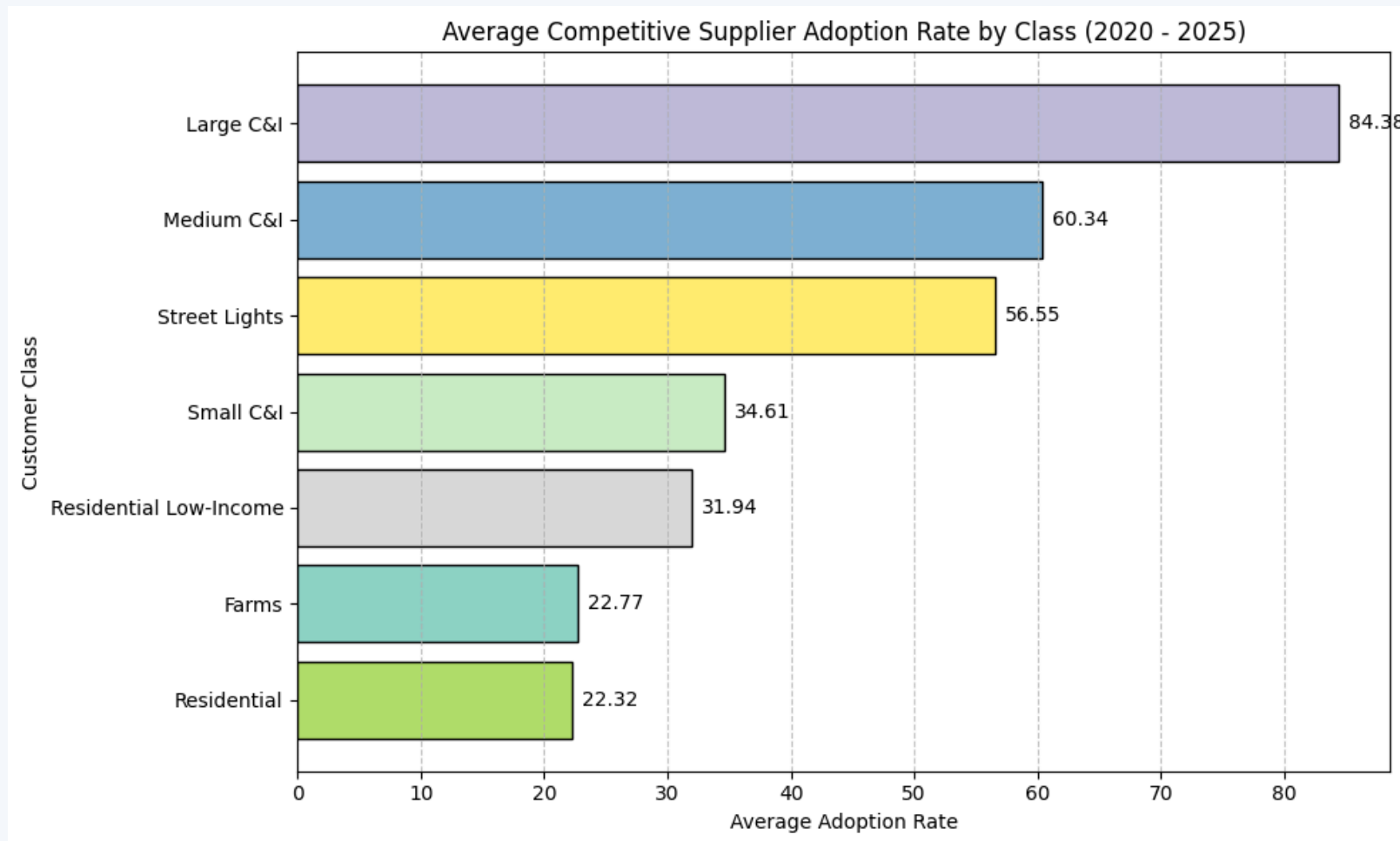


Residential consumption shows the strongest seasonal swing (winter heating and summer cooling peaks). Large C&I consumption is more stable but still shows mild seasonal variation. The time series confirms that seasonal patterns differ significantly by class.

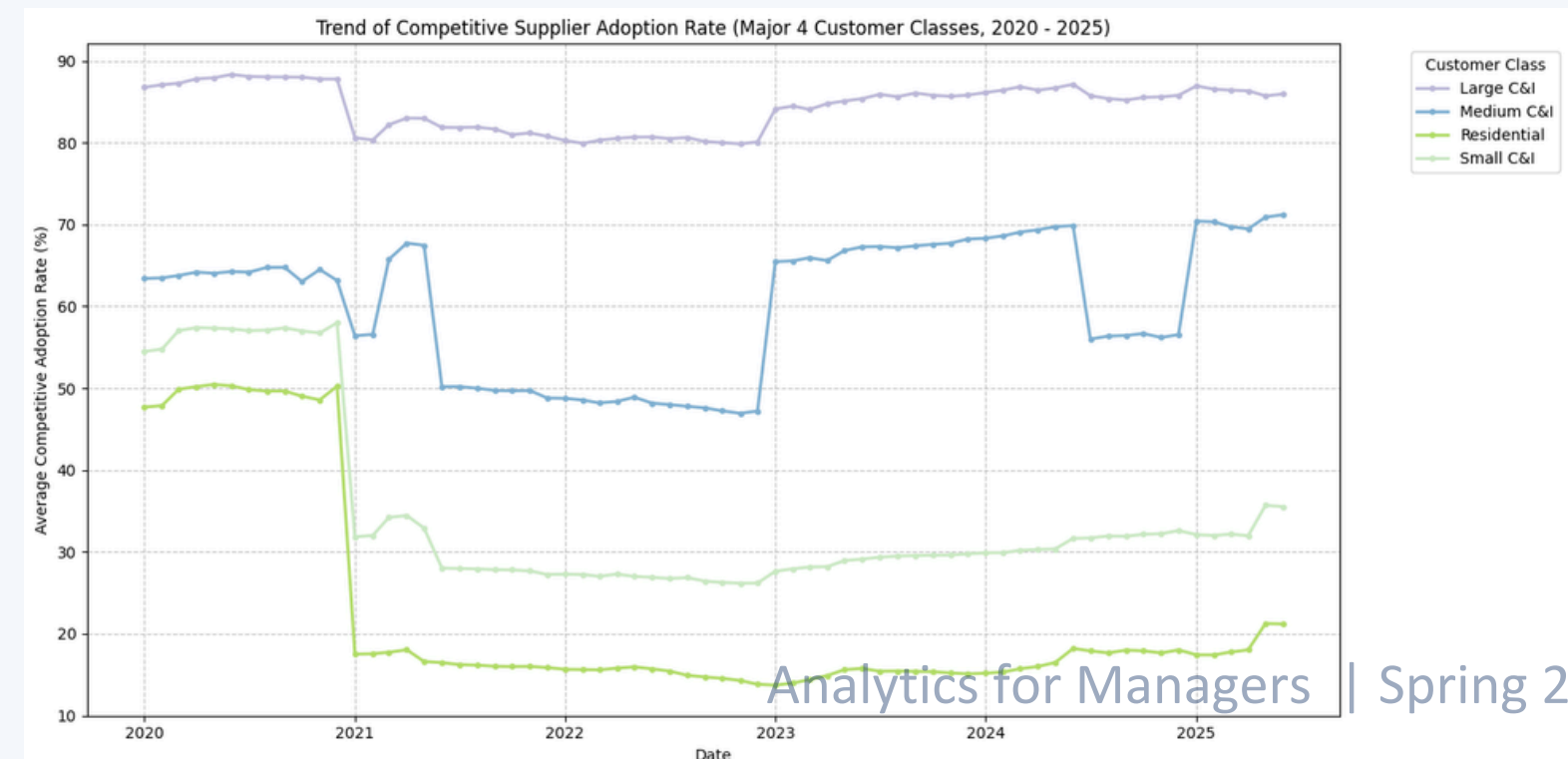
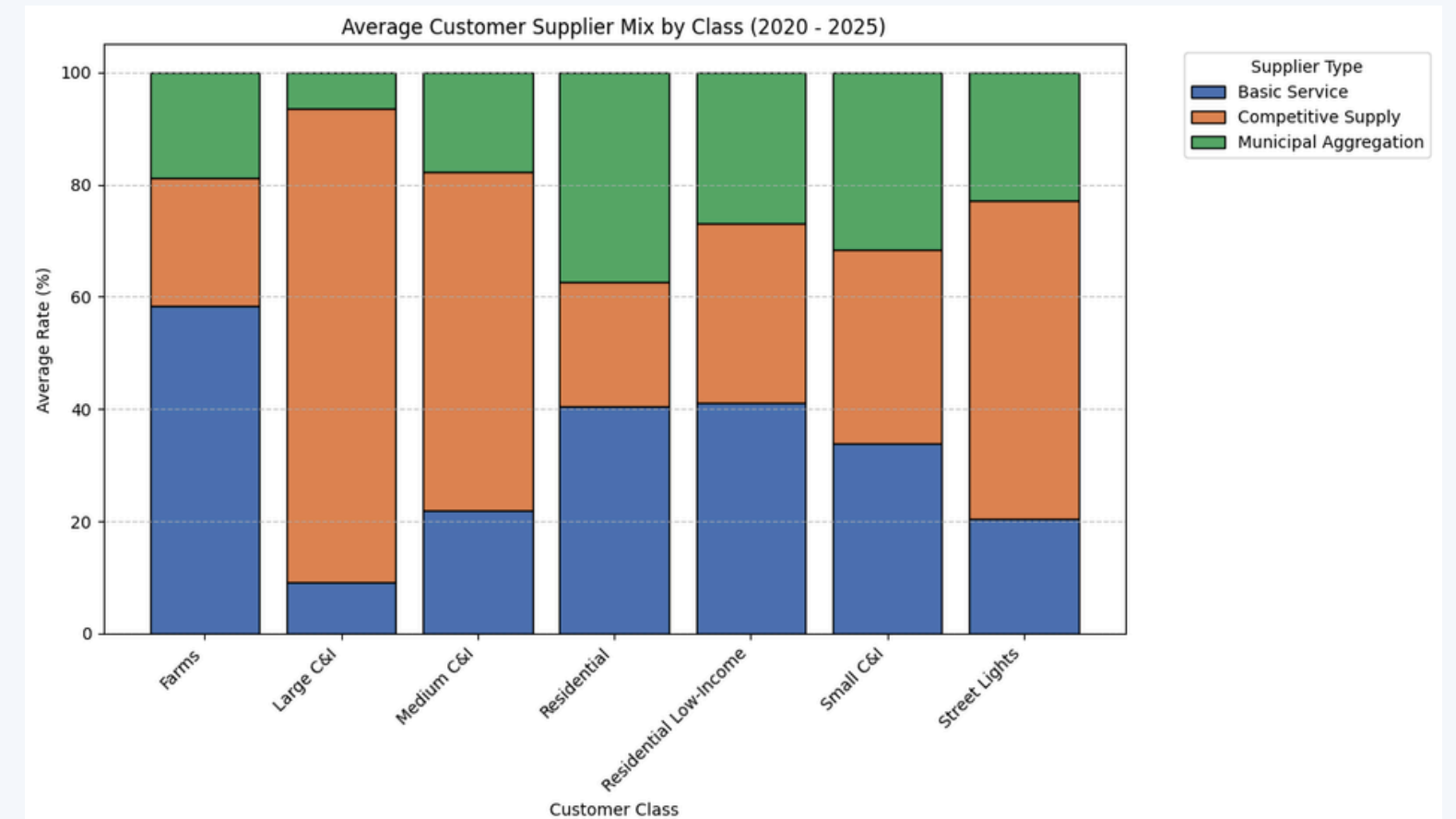


When aggregated across all classes, system-wide electricity demand reaches its highest levels in **July to August** due to cooling needs and shows a secondary peak in **January to February** associated with heating; demand is lowest during the shoulder months of April to May and October to November

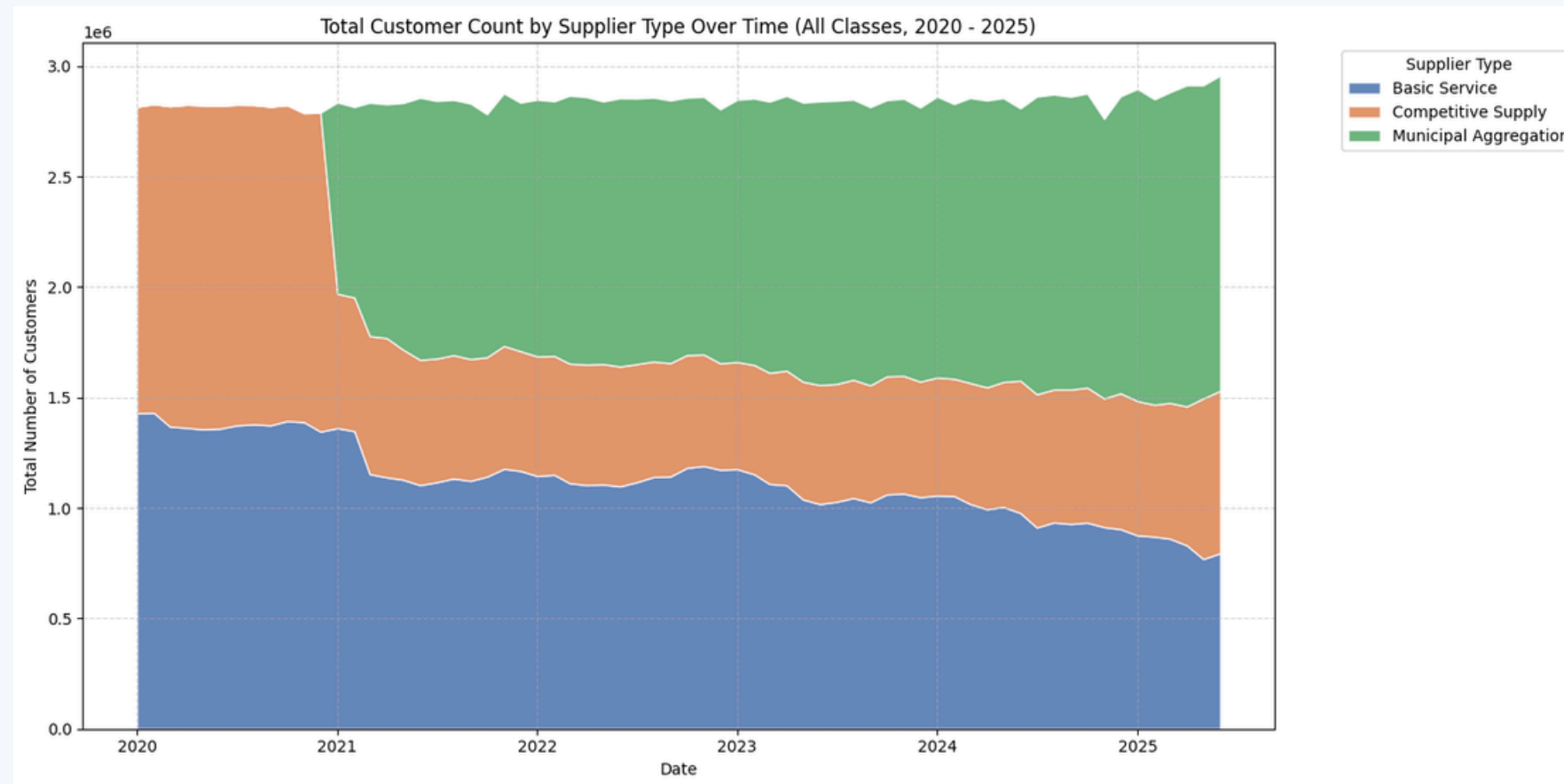
Key Findings: Competitive supplier market is driven by C&I segment



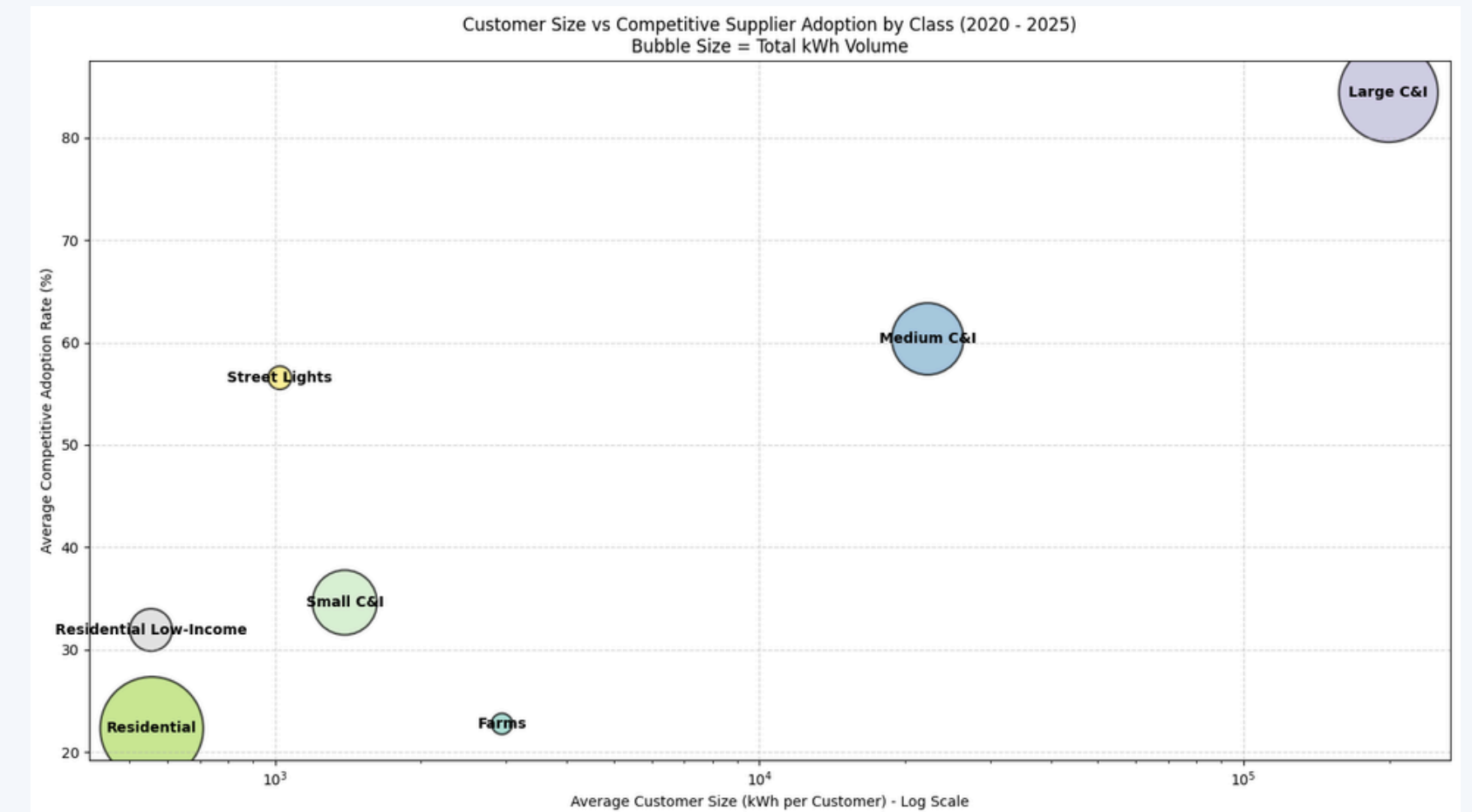
- A significant disruption in the competitive supplier participation across all customer segments in 2021.
- Large C&I customers have the highest engagement at 84.4% competitive adoption rate, indicating majority have moved away from basic utility service.
- Residential customers have the lowest switching rate at 22.3 percent, but high participation for Municipal Aggregation programs
- Farms are the most basic service-dependent class, with roughly 59% still on basic service.



Key Findings: Consumption volume is a key driver of competitive supplier adoption across customer classes.



- Residential customers switched from basic utility services and competitive suppliers to Municipal aggregation, which became a major segment by 2025.
- The steady yet consistent decline of the basic service suggests a gradual shift away from incumbent utility providers to municipal utility aggregation rather than the competitive suppliers.



- Larger users, particularly large C&I accounts, exhibit a significantly higher propensity to engage with competitive suppliers, confirming that consumption volume is a key driver of competitive supplier adoption across customer classes.

Key Findings: Supplier Type is strongly predicted by Historical patterns

```
Classification Report:
              precision    recall  f1-score   support

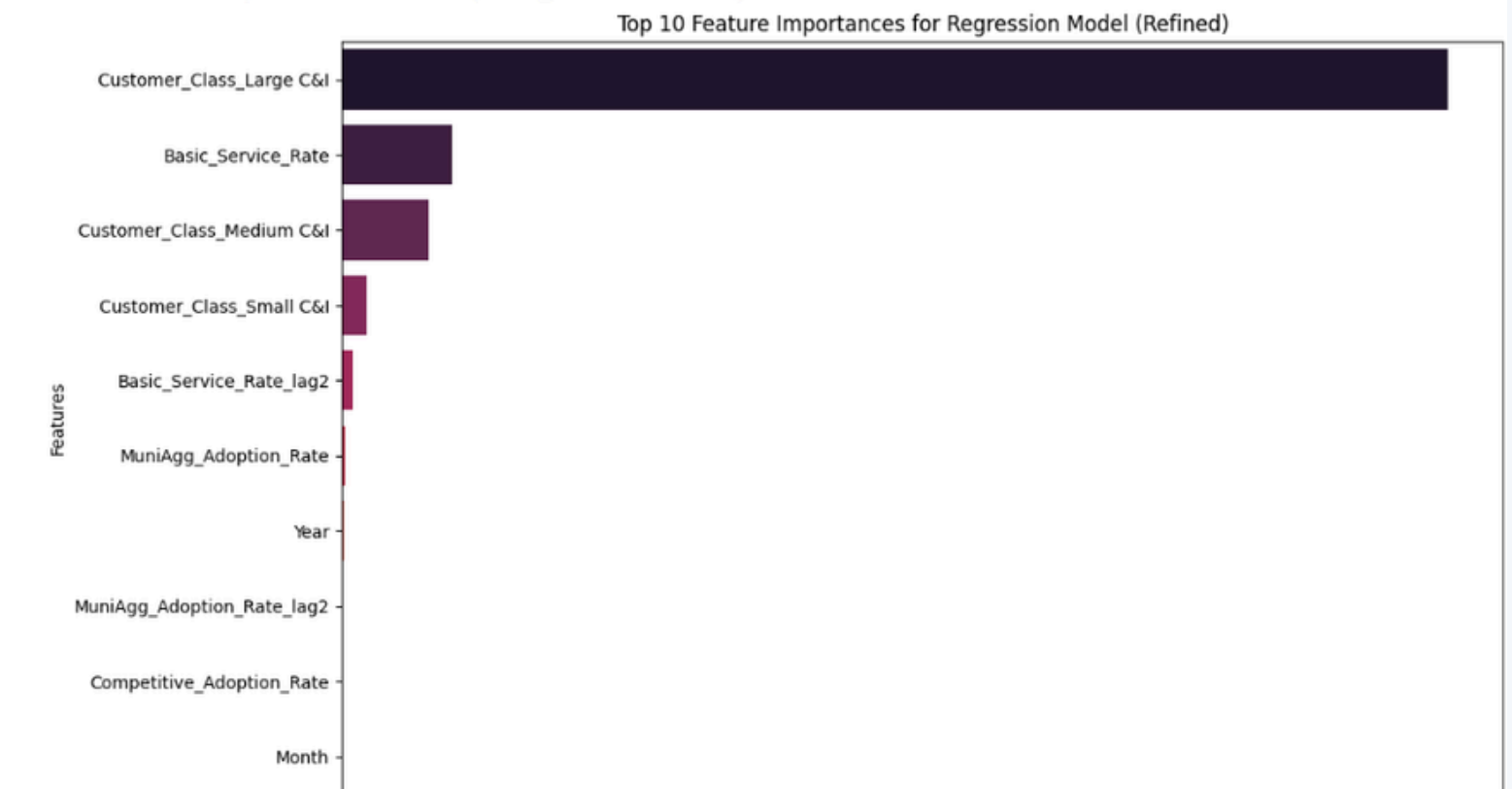
     Basic           0.00      0.00      0.00         0
  Competitive       1.00      0.95      0.98        65
    Municipal       1.00      1.00      1.00        23

 accuracy              0.97         88
  macro avg           0.67      0.65      0.66         88
 weighted avg         1.00      0.97      0.98         88
```

```
Top 10 most important features for classification:
MuniAgg_Adoption_Rate_lag2          0.374806
Competitive_Adoption_Rate_lag1      0.354018
Basic_Service_Rate_lag1              0.074239
Basic_Service_Rate_lag2              0.063126
MuniAgg_Adoption_Rate_lag1          0.062613
Month                               0.016021
Customer_Class_Small C&I            0.015938
Customer_Class_Residential Low-Income 0.014329
Year                                0.013266
Customer_Class_Residential           0.008812
dtype: float32
```

- The classification model confirmed that historical behavior and patterns strongly predict current supplier type, with past competitive adoption rate and past municipal aggregation rate being the strongest. However, the model's ability to predict basic service could not be validated due to class imbalance in the test set.

```
--- Regression Model Summary Statistics ---
Regression R2 Score (original scale): -2500.7923
Mean Squared Error (original scale): 405019637.66
Root Mean Squared Error (original scale): 20125.10
```



- The regression model failed to produce reliable consumption estimates, as indicated by an R^2 score of -2500. As a result, the relationship between competitive supplier adoption and average consumption per customer could not be meaningfully established. The model requires significant rework before its findings can be used for decision-making.

Strategic Takeaways

- 1 Customer class is the central driver of both electricity consumption and competitive supplier adoption in Massachusetts.
- 2 Large C&I customers, though few in number, consume nearly as much electricity as the entire residential sector while maintaining consistently high competitive adoption.
- 3 Retail competition works well for high-volume C&I users; but residential and small C&I customers requires community-level mechanisms for engagement. One-size policy does not fit all.
- 4 A strong positive relationship between average monthly consumption and the propensity to leave basic service; alongside system-wide seasonal consumption patterns.
- 5 The classification model achieved 97% accuracy in predicting dominant supply type using only lagged historical adoption rates, confirming the predictive power of recent participation trends.
- 6 The regression model performed poorly in forecasting average kWh per customer due to extreme variance from large industrial accounts.

Recommendations: Improved competitive supplier participation

Defend and Deepen Large C&I Retention

Shift performance metrics from acquisition to retention, using lagged participation data from the classification model to flag disengaging accounts early, given that this segment drives 31.7% of total consumption, and losing even a few accounts creates significant procurement and revenue.

Develop targeted residential re-engagement strategies

Develop residential-specific product offerings to address the low per-customer consumption profile, such as community-based pricing, bundled services, or simplified fixed-rate contracts that raise switching costs and make competitive market participation viable.

Prioritize Medium C&I as a growth market

Target the 39.7% still on basic service by using DOER data to identify the lowest-penetration utility territories first, and offer standardized fixed-price products that are easier to hedge given the segment's relatively stable seasonal demand profile.

Investigate and address the causes of the 2021 market disruption

A review of the decline in the competitive market adoption across all classes in 2021, leading to a disproportionate and long-lasting consequential impact on the residential class, will reveal where intervention for resilience and recovery is needed.

Thank You!

